

3. Sums of Independent Random Variables

Suppose that $X_1, X_2, \dots, X_n$ are independent random variables, and let $Y = \sum_i X_i$. Then,

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$$

for all t such that all $M_{X_i}(t)$ is finite.

Exercise: Prove the above result.

$$M_Y(t) = E(e^{tY}) = E(e^{t(X_1 + X_2 + \dots + X_n)})$$

$$= E(e^{tX_1} e^{tX_2} \dots e^{tX_n})$$

$$= \prod_{i=1}^n E(e^{tX_i}) \quad \text{since } X_i \text{ are independent}$$

$$= \prod_{i=1}^n M_{X_i}(t)$$

Exercise: Show that the sum of independent Poisson random variables also has the Poisson distribution.

$$X_1 \sim \text{Poisson}(\lambda_1) \quad X_2 \sim \text{Poisson}(\lambda_2), \quad X_1, X_2 \text{ independent}$$

$$Y = X_1 + X_2$$

$$M_Y(t) = M_{X_1}(t) M_{X_2}(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$$

$$\text{So, } Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

Exercise: Under what condition(s) does the sum of Gamma random variables still have a Gamma distribution?

$$X_i \sim \text{Gamma}(\alpha_i, \beta_i) \quad i=1, 2, \dots, n$$

$$M_{X_i}(t) = \left(\frac{\beta_i}{\beta_i - t} \right)^{\alpha_i}, \quad t < \beta_i$$

Assume $X_1, X_2, \dots, X_n$ independent, $Y = \sum X_i$

$$M_Y(t) = \prod_{i=1}^n \left(\frac{\beta_i}{\beta_i - t} \right)^{\alpha_i}$$

In order for this to remain Gamma,

all β_i must be equal. So need $\beta_1 = \beta_2 = \dots = \beta_n = \beta$

$$\text{Then, } M_Y(t) = \prod_{i=1}^n \left(\frac{\beta}{\beta - t} \right)^{\alpha_i} = \left(\frac{\beta}{\beta - t} \right)^{\sum \alpha_i}$$

So, no restrictions on α_i needed. $Y \sim \text{Gamma}(\beta, \sum \alpha_i)$

Comments: ① If $X_1, X_2, \dots, X_n$ iid Exponential(λ),

then $\sum X_i \sim \text{Gamma}(n, \lambda)$

② If $X \sim \text{Gamma}(\alpha, \beta)$, then $kX \sim \text{Gamma}(\alpha, \beta/k)$
for $k > 0$.

4. Establishing Convergence in Distribution

In some situations it will be important to formalize the notion of "the distribution of $X_1, X_2, \dots$ converges to the distribution of X ."

In such cases, we would write $X_n \xrightarrow{D} X$, and say that " X_n converges in distribution to X ." Does not imply that $X_n(\omega) \rightarrow X(\omega)$ for all ω .

The formal definition is as follows: X_n **converges in distribution** to X if $F_{X_n}(x) \rightarrow F_X(x)$ as $n \rightarrow \infty$ at all x where F_X is continuous.

In practical terms, the value of knowing $X_n \xrightarrow{D} X$ is that (for large n), the distribution of ~~X_n~~ can be taken as an approximation to the distribution of ~~X_n~~ . The **central limit theorem** is an important example.

The useful result: $X_n \xrightarrow{D} X$ if $M_{X_n}(t) \rightarrow M_X(t)$ as $n \rightarrow \infty$ for all $-\epsilon < t < \epsilon$ for some $\epsilon > 0$.

1 **Exercise:** Suppose that X equals zero with probability one and X_n
2 equals $1/n$ with probability one. Does $X_n \xrightarrow{D} X$?

3 Heuristically, it seems the dist. of X_n does
4 get closer to the dist. of X as n grows.

5
6
$$F_X(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

7

8
9
$$F_{X_n}(x) = \begin{cases} 1, & x \geq 1/n \\ 0, & x < 1/n \end{cases}$$

10

11
12
$$0 = F_{X_n}(0) \not\rightarrow F_X(0) = 1$$

13

14 Note that the CDF of X is not continuous
15 at 0. So, the fact that $F_{X_n}(0) \not\rightarrow F_X(0)$
16 does not mean that $X_n \not\xrightarrow{D} X$.
17

18 In fact, $F_{X_n}(x) \rightarrow F_X(x)$ at all $x \neq 0$
19 So, $X_n \xrightarrow{D} X$
20

Exercise: Suppose that X_n has the Binomial($n, \lambda/n$) distribution for $\lambda > 0$. Let X have the Poisson(λ) distribution. Show that $X_n \xrightarrow{D} X$.

$$\begin{aligned} M_{X_n}(t) &= \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^t\right)^n \\ &= \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \end{aligned}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{X_n}(t) &= \lim_{n \rightarrow \infty} \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \\ &= e^{\lambda(e^t - 1)} \end{aligned}$$

since $e^a = \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n$

$$= M_X(t)$$

$$\text{So, } X_n \xrightarrow{D} X$$

The Normal Distribution and the MGF

If X has the normal(μ, σ^2) distribution (with pdf given previously in the notes), then

$$M_X(t) = e^{t\mu + t^2\sigma^2/2}, \quad -\infty < t < \infty$$

The following properties of the normal(μ, σ^2) distribution are easy to prove once we know the MGF and its properties:

1. $E(X) = \mu$

2. $E(X^2) = \mu^2 + \sigma^2$, and hence $V(X) = \sigma^2$

3. If $Y = aX + b$, with $a \neq 0$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.

4. The linear combination of independent normal random variables also has the normal distribution. In particular,

(a) If $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\sum_i X_i \sim N(n\mu, n\sigma^2)$.

(b) If $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.

5. The Central Limit Theorem: Suppose $X_1, X_2, \dots$ are i.i.d. and

$E(X_i) = \mu$ and $V(X_i) = \sigma^2$ both exist and are finite, then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{D} Z$$

where $\bar{X}_n = \sum_{i=1}^n X_i/n$ and $Z \sim N(0, 1)$.

From DeGroot and Schervish,
Fourth Ed.

and

$$\begin{aligned}\sum_{i=1}^{99} p_i(1-p_i) &= \frac{1}{100} \sum_{i=1}^{99} i - \frac{1}{(100)^2} \sum_{i=1}^{99} i^2 \\ &= 49.5 - \frac{1}{(100)^2} \cdot \frac{(99)(100)(199)}{6} = 16.665.\end{aligned}$$

It follows from the central limit theorem that the distribution of the total number of questions that are answered correctly, which is $\sum_{i=1}^{99} X_i$, will be approximately the normal distribution with mean 49.5 and standard deviation $(16.665)^{1/2} = 4.08$. Therefore, the distribution of the variable

$$Z = \frac{\sum_{i=1}^n X_i - 49.5}{4.08}$$

will be approximately the standard normal distribution. It follows that

$$\Pr\left(\sum_{i=1}^n X_i \geq 60\right) = \Pr(Z \geq 2.5735) \simeq 1 - \Phi(2.5735) = 0.0050.$$

Outline of Proof of Central Limit Theorem

Convergence of the Moment Generating Functions Moment generating functions are important in the study of convergence in distribution because of the following theorem, the proof of which is too advanced to be presented here.

Theorem 6.3.5

Let $X_1, X_2, \dots$ be a sequence of random variables. For $n = 1, 2, \dots$, let F_n denote the c.d.f. of X_n , and let ψ_n denote the m.g.f. of X_n .

Also, let X^* denote another random variable with c.d.f. F^* and m.g.f. ψ^* . Suppose that the m.g.f.'s ψ_n and ψ^* exist ($n = 1, 2, \dots$). If $\lim_{n \rightarrow \infty} \psi_n(t) = \psi^*(t)$ for all values of t in some interval around the point $t = 0$, then the sequence $X_1, X_2, \dots$ converges in distribution to X^* . ■

In other words, the sequence of c.d.f.'s $F_1, F_2, \dots$ must converge to the c.d.f. F^* if the corresponding sequence of m.g.f.'s $\psi_1, \psi_2, \dots$ converges to the m.g.f. ψ^* .

Outline of the Proof of Theorem 5.7.1 We are now ready to outline a proof of Theorem 6.3.1, which is the central limit theorem of Lindeberg and Lévy. We shall assume that the variables $X_1, \dots, X_n$ form a random sample of size n from a distribution with mean μ and variance σ^2 . We shall also assume, for convenience, that the m.g.f. of this distribution exists, although the central limit theorem is true even without this assumption.

For $i = 1, \dots, n$, let $Y_i = (X_i - \mu)/\sigma$. Then the random variables $Y_1, \dots, Y_n$ are i.i.d., and each has mean 0 and variance 1. Furthermore, let

$$Z_n = \frac{n^{1/2}(\bar{X}_n - \mu)}{\sigma} = \frac{1}{n^{1/2}} \sum_{i=1}^n Y_i.$$

We shall show that Z_n converges in distribution to a random variable having the standard normal distribution, as indicated in Eq. (6.3.1), by showing that the m.g.f. of Z_n converges to the m.g.f. of the standard normal distribution.

If $\psi(t)$ denotes the m.g.f. of each random variable Y_i ($i = 1, \dots, n$), then it follows from Theorem 4.4.4 that the m.g.f. of the sum $\sum_{i=1}^n Y_i$ will be $[\psi(t)]^n$. Also, it follows from Theorem 4.4.3 that the m.g.f. $\zeta_n(t)$ of Z_n will be

$$\zeta_n(t) = \left[\psi\left(\frac{t}{n^{1/2}}\right) \right]^n.$$

In this problem, $\psi'(0) = E(Y_i) = 0$ and $\psi''(0) = E(Y_i^2) = 1$. Therefore, the Taylor series expansion of $\psi(t)$ about the point $t = 0$ has the following form:

$$\begin{aligned} \psi(t) &= \psi(0) + t\psi'(0) + \frac{t^2}{2!}\psi''(0) + \frac{t^3}{3!}\psi'''(0) + \dots \\ &= 1 + \frac{t^2}{2} + \frac{t^3}{3!}\psi'''(0) + \dots \end{aligned}$$

Also,

$$\zeta_n(t) = \left[1 + \frac{t^2}{2n} + \frac{t^3\psi'''(0)}{3!n^{3/2}} + \dots \right]^n. \quad (6.3.15)$$

Apply Theorem 5.3.3 with $1 + a_n/n$ equal to the expression inside brackets in (6.3.15) and $c_n = n$. Since

$$\lim_{n \rightarrow \infty} \left[\frac{t^2}{2} + \frac{t^3\psi'''(0)}{3!n^{1/2}} + \dots \right] = \frac{t^2}{2},$$

it follows that

$$\lim_{n \rightarrow \infty} \zeta_n(t) = \exp\left(\frac{1}{2}t^2\right). \quad (6.3.16)$$

Since the right side of Eq. (6.3.16) is the m.g.f. of the standard normal distribution, it follows from Theorem 6.3.5 that the asymptotic distribution of Z_n must be the standard normal distribution.

An outline of the proof of the central limit theorem of Liapounov can also be given by proceeding along similar lines, but we shall not consider this problem further here.

Summary

Two versions of the central limit theorem were given. They conclude that the distribution of the average of a large number of independent random variables is close to a normal distribution. One theorem requires that the random variables all have the same distribution with finite variance. The other theorem does not require that the random variables be identically distributed, but instead requires that their third moments exist and satisfy condition (6.3.9). The delta method lets us find the approximate distribution of a smooth function of a sample average.

Exercise: What is the practical value of the CLT?

When n is "large", $\bar{X}_n \approx N(\mu, \sigma^2/n)$

Similar to saying $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$

Exercise: Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables, each with the Exponential(1) distribution. What is the distribution of $\bar{X}$?

$$E(X_i) = 1 = \mu \quad V(X_i) = 1 = \sigma^2$$

$$\sum_{i=1}^n X_i \sim \text{Gamma}(n, 1)$$

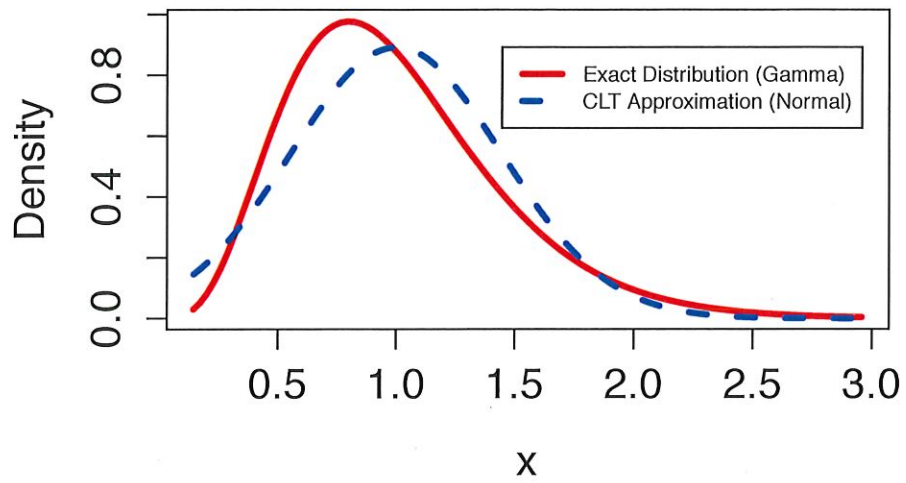
$$\bar{X}_n = \frac{\sum X_i}{n} \sim \text{Gamma}(n, n)$$

The CLT says that

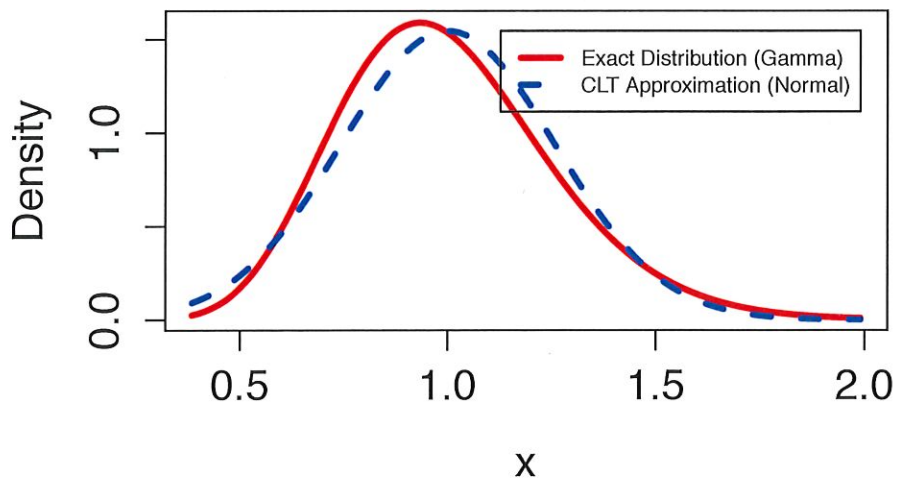
$$\bar{X}_n \approx N(\mu, \sigma^2/n), N(1, 1/n)$$

for n "large"

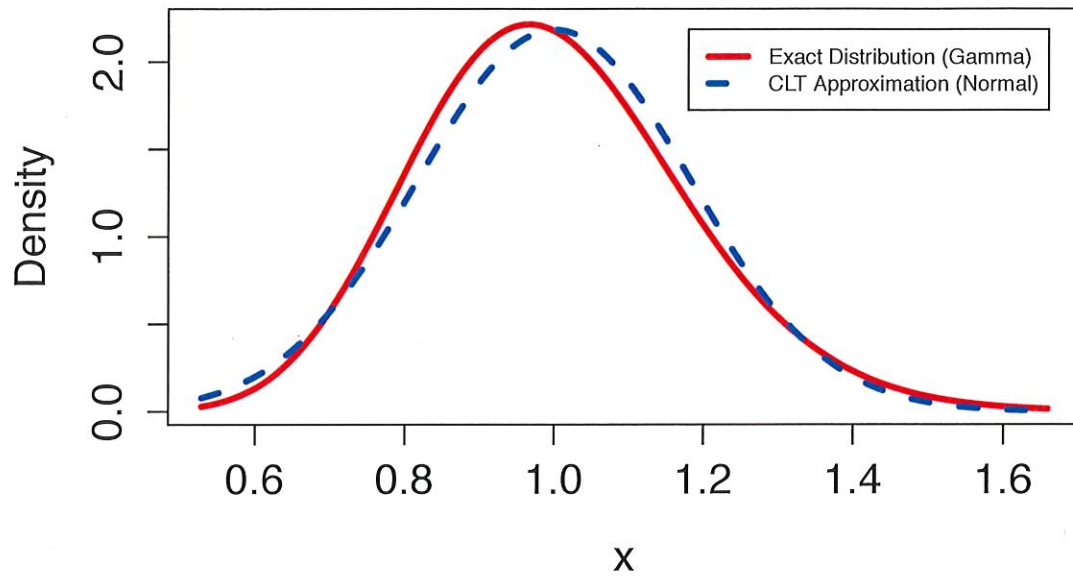
With $n = 5$



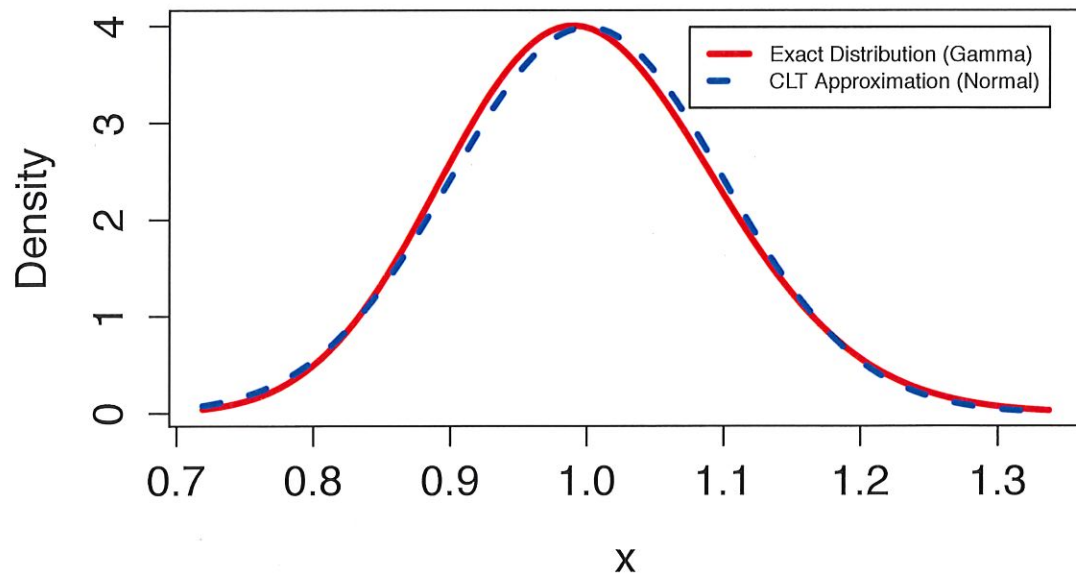
With $n = 15$



With $n = 30$



With $n = 100$



Exercise: Suppose that on any day the price of a particular stock either increases by 2% (with probability 0.5) or decreases by 1% (with probability 0.5). Each day's change is independent of the other days. What is the probability that the stock's price is more than twice its current price 100 trading days from now?

Let $P_{100} = X_1 X_2 \dots X_{100}$ where X_i are iid such that

$$X_i = \begin{cases} 1.02 & \text{w.p. } 1/2 \\ 0.99 & \text{w.p. } 1/2 \end{cases}$$

P_{100} = return after 100 days

$$\{\text{stock price doubles after 100 days}\} = \{P_{100} > 2\}$$

$$\log P_{100} = \sum_{i=1}^{100} \log X_i = \sum_{i=1}^{100} Y_i$$

$$Y_i = \begin{cases} \log(1.02) & \text{w.p. } 1/2 \\ \log(0.99) & \text{w.p. } 1/2 \end{cases}$$

and Y_i are iid

$$\mu = E(Y_i) = \log(1.02)(1/2) + \log(0.99)(1/2) = 0.00488$$

$$\sigma^2 = V(Y_i) = E(Y_i^2) - E(Y_i)^2 = 0.000223$$

CLT says that $\bar{Y}_n \approx N(\mu, \sigma^2/n)$

Note that $P_{100} > 2$ iff $\log(P_{100}) > \log(2)$
iff $\sum Y_i > \log(2)$
iff $\bar{Y} > \log(2)/100$

$$\text{So, } P(P_{100} > 2) = P(\bar{Y} > \frac{\log(2)}{100})$$

by CLT

$$\approx P\left(Z > \frac{\frac{\log(2)}{100} - \mu}{\frac{\sigma}{10}}\right)$$

1.37

where $Z \sim N(0, 1)$

$$= 1 - \Phi(1.37)$$

$$\approx 0.08$$

1 The Delta Method

2 A formal statement of the Delta Method is as follows:

3 Assume that $Y_1, Y_2, \dots, Y_n$ are such that

$$4 \quad \frac{Y_n - \mu}{\sigma_n} \xrightarrow{D} Z$$

5 where $Z \sim N(0, 1)$, μ is a constant, and σ_n is such that $\sigma_n > 0$
6 and $\sigma_n \downarrow 0$ as $n \rightarrow \infty$. Then, assuming $g(\cdot)$ is a function that
7 is differentiable at μ and $g'(\mu) \neq 0$, it holds that

$$8 \quad \frac{g(Y_n) - g(\mu)}{|g'(\mu)|\sigma_n} \xrightarrow{D} Z.$$

9 A more heuristic, but useful, version is as follows:

10 If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \downarrow 0$ as $n \rightarrow \infty$, then
11 $g(Y_n)$ is approximately

$$12 \quad N(g(\mu), (g'(\mu))^2 \sigma_n^2)$$

13 assuming that $g(\cdot)$ is a function that is differentiable at μ and
14 $g'(\mu) \neq 0$.

15 This is often (but not always) applied with $Y_n = \bar{X}_n$ and $\sigma_n^2 = \sigma^2/n$
16 where $\sigma^2 > 0$.

Exercise: Suppose that $X_1, X_2, \dots, X_{50}$ are i.i.d. random variables with the Exponential($\lambda = 2$) distribution. One may hope that $1/\bar{X}$ is a good estimator of λ (imagining a case where λ is not known). What is the probability that $1/\bar{X}$ is within 0.1 of λ ?

$$E(X_i) = 1/\lambda$$

$$\bar{X} \approx E(X_i)$$

$$\lambda \approx 1/\bar{X}$$

$$\mu = E(X_i) = 1/2$$

$$\sigma^2 = V(X_i) = 1/4$$

By the CLT, $\bar{X} \approx N(1/2, 1/(50 \times 4))$
being more general $\bar{X} \approx N(1/\lambda, 1/n\lambda^2)$

But want the (approx.) dist. of $1/\bar{X}$
Use Δ -method with $g(\bar{X}) = 1/\bar{X}$
i.e. $g(x) = 1/x$

So, $g'(x) = -1/x^2$ and by the Δ -method,
 $g(\bar{X}) = 1/\bar{X}$ is approx. Normal
with mean $g(\mu) = 1/2 = \lambda$
and variance

$$\left(g'(\mu)\right)^2 \frac{\sigma^2}{n} = \left(g'\left(\frac{1}{\lambda}\right)\right)^2 \left(\frac{1}{n\lambda^2}\right)$$

$$= \left(-\frac{1}{(\frac{1}{\lambda})^2}\right)^2 \left(\frac{1}{n\lambda^2}\right) = \lambda^2/n$$

Claim that by the Δ -method,

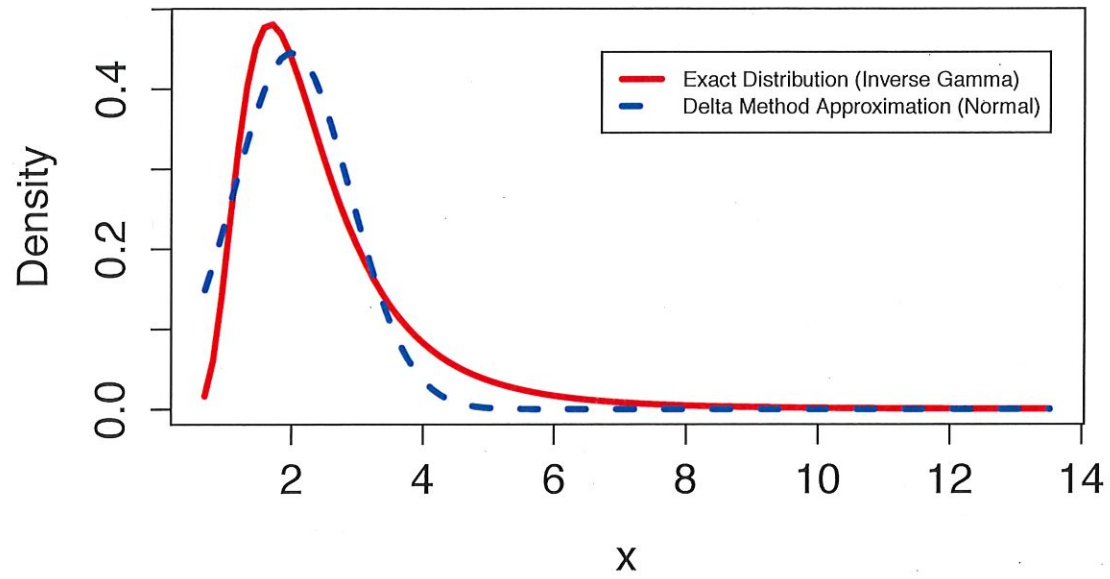
$$\frac{1}{\bar{X}} \approx N\left(\bar{\lambda}, \frac{\bar{\lambda}^2}{n}\right)$$

In fact, $\bar{X} \sim \text{Gamma}(n, n\bar{\lambda})$

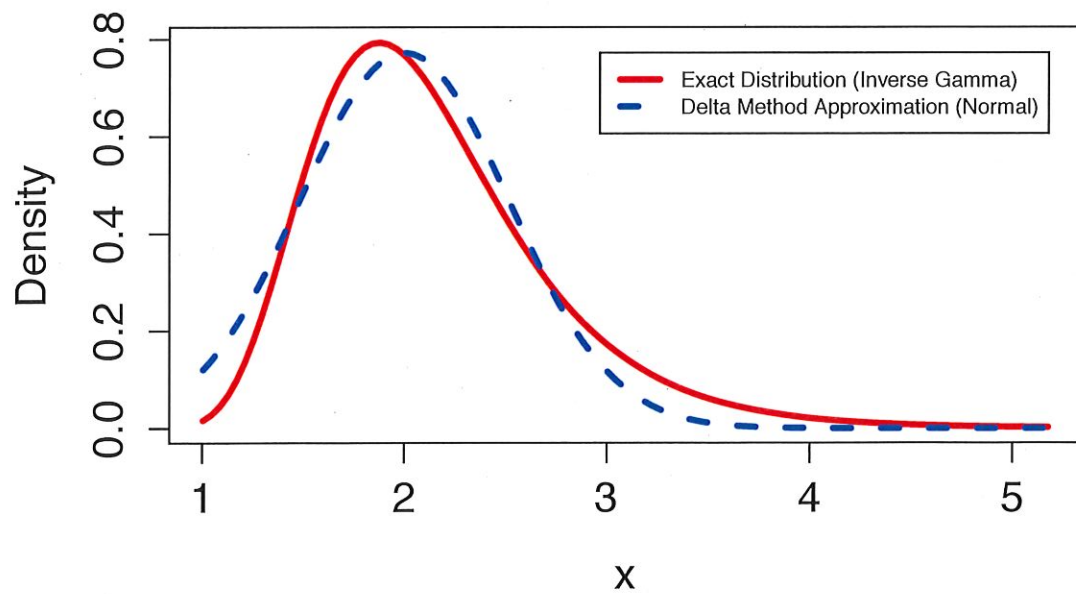
and $\frac{1}{\bar{X}} \sim \text{Inverse Gamma}(n, n\bar{\lambda})$

Plug in $\bar{\lambda}=2$ and compare dsts. for
different choices of n

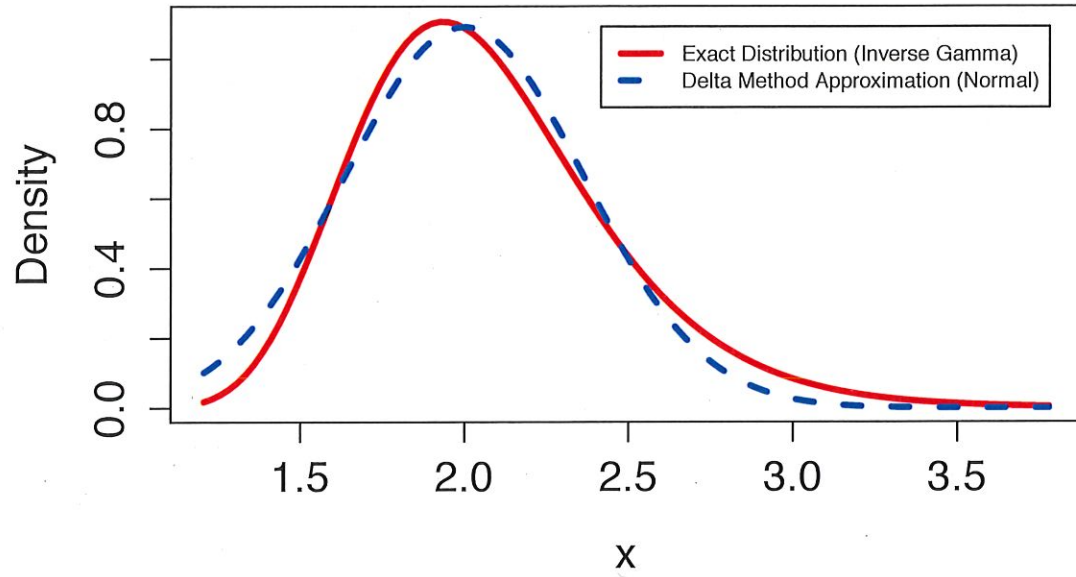
With $n = 5$



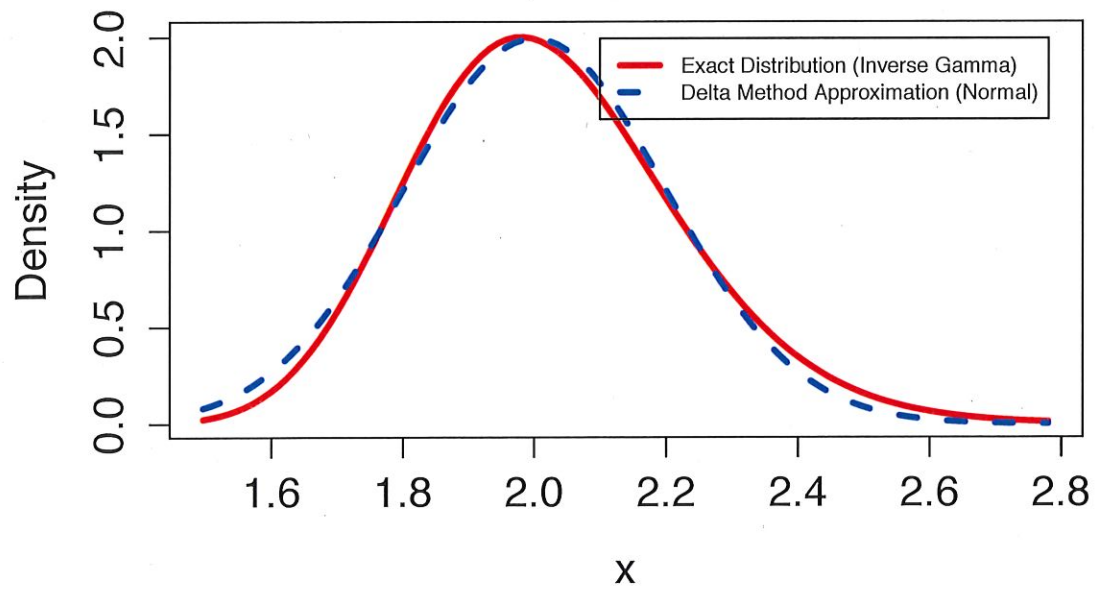
With $n = 15$



With $n = 30$



With $n = 100$



$$\text{If } \mu=2, n=50, \bar{X} \approx N(2, 4/50)$$

$$P(\bar{X} \text{ is within } 0.1 \text{ of } \mu=2)$$

$$= P(1.9 \leq \bar{X} \leq 2.1)$$

$$\text{In R} \quad = \text{pnorm}(2.1, 2, \sqrt{4/50}) - \text{pnorm}(1.9, 2, \sqrt{4/50})$$

$$= P\left(\frac{1.9-2}{\sqrt{4/50}} \leq \frac{\bar{X}-2}{\sqrt{4/50}} \leq \frac{2.1-2}{\sqrt{4/50}}\right)$$

$$= P\left(\frac{1.9-2}{\sqrt{4/50}} \leq Z \leq \frac{2.1-2}{\sqrt{4/50}}\right)$$

$$= \Phi\left(\frac{2.1-2}{\sqrt{4/50}}\right) - \Phi\left(\frac{1.9-2}{\sqrt{4/50}}\right) = 0.276$$

$\hat{\theta}$ is the MLE for θ

Theory: $\hat{\theta} \approx N(\theta, \frac{1}{g(\theta)})$

Δ method $g(\hat{\theta}) \approx N(\theta, \frac{1}{g(\theta)})$

$$X \sim \text{Exp}(\lambda) \quad P(X > a) = e^{-\lambda a} = g(a)$$